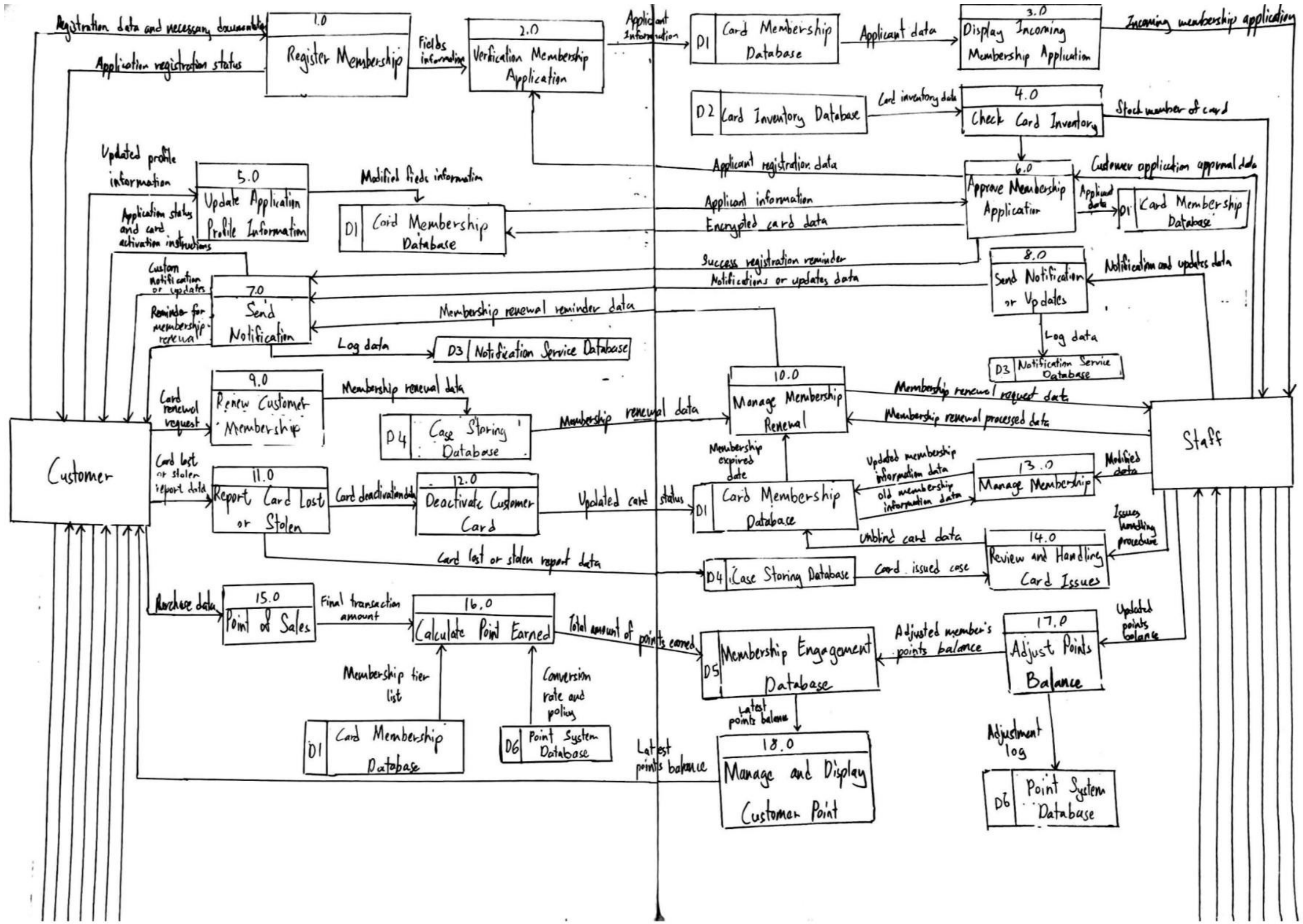
2.1 Membership Card Management System Data Flow Diagram Level 0



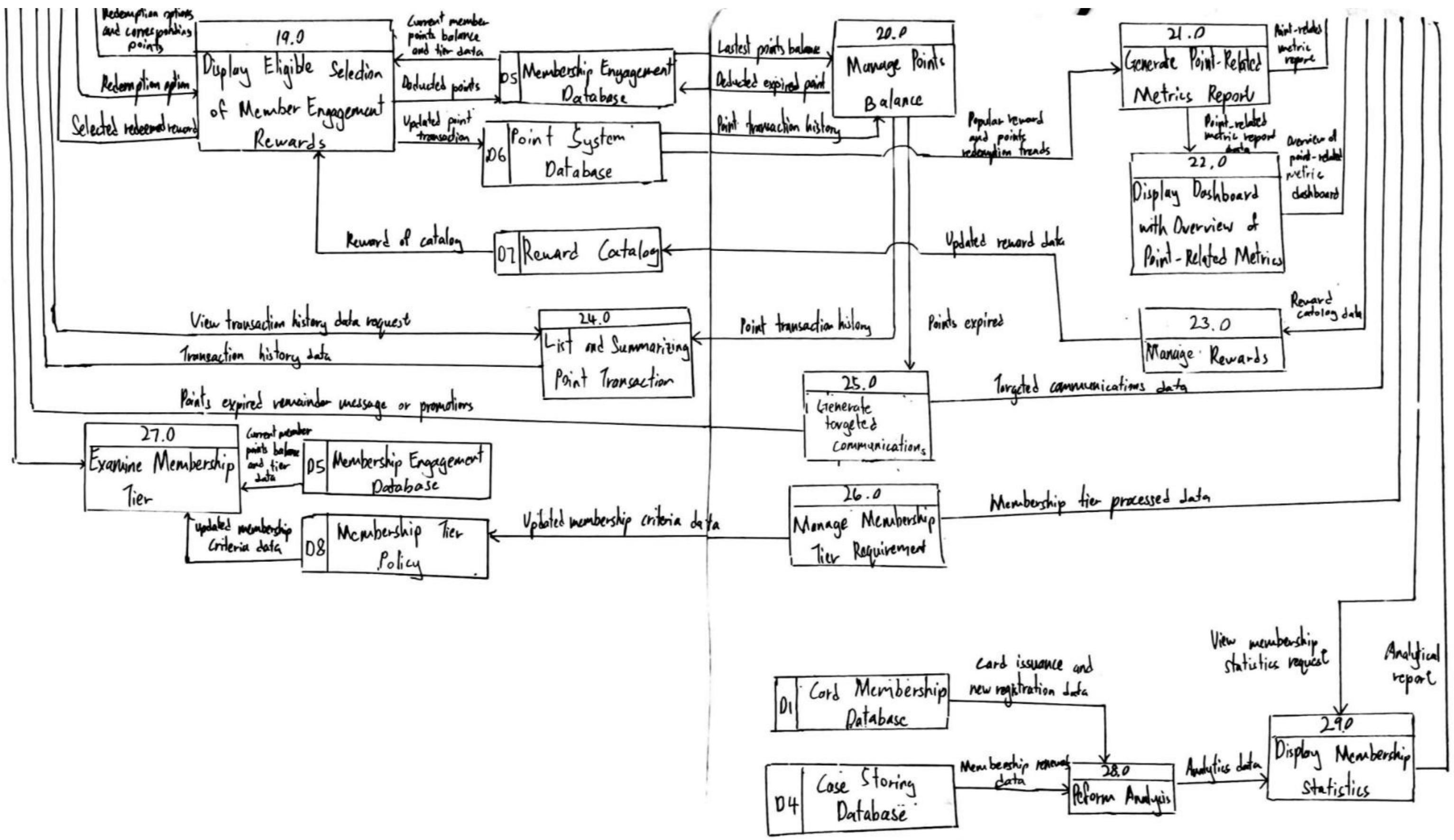


Figure 2.1.1: Hand Drawn Membership Card Management System Data Flow Diagram Level 0

2.2 System’s Functionalities for The Membership Card Management System

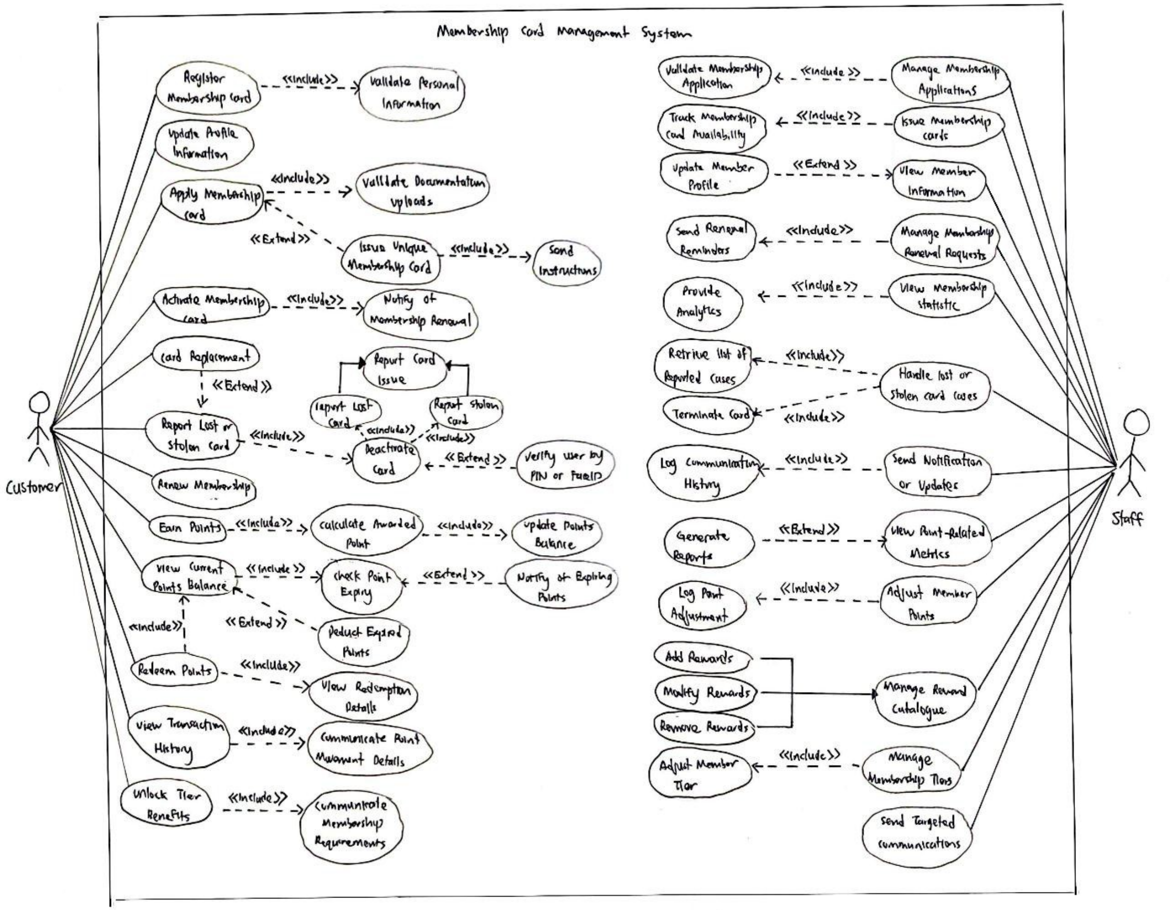


Figure 2.2.1: Hand Drawn System’s Functionalities for The Membership Card Management System Use Case Diagram

2.3 DFD-0 and Use Case Diagrams Using Software

2.3.1 DFD-0

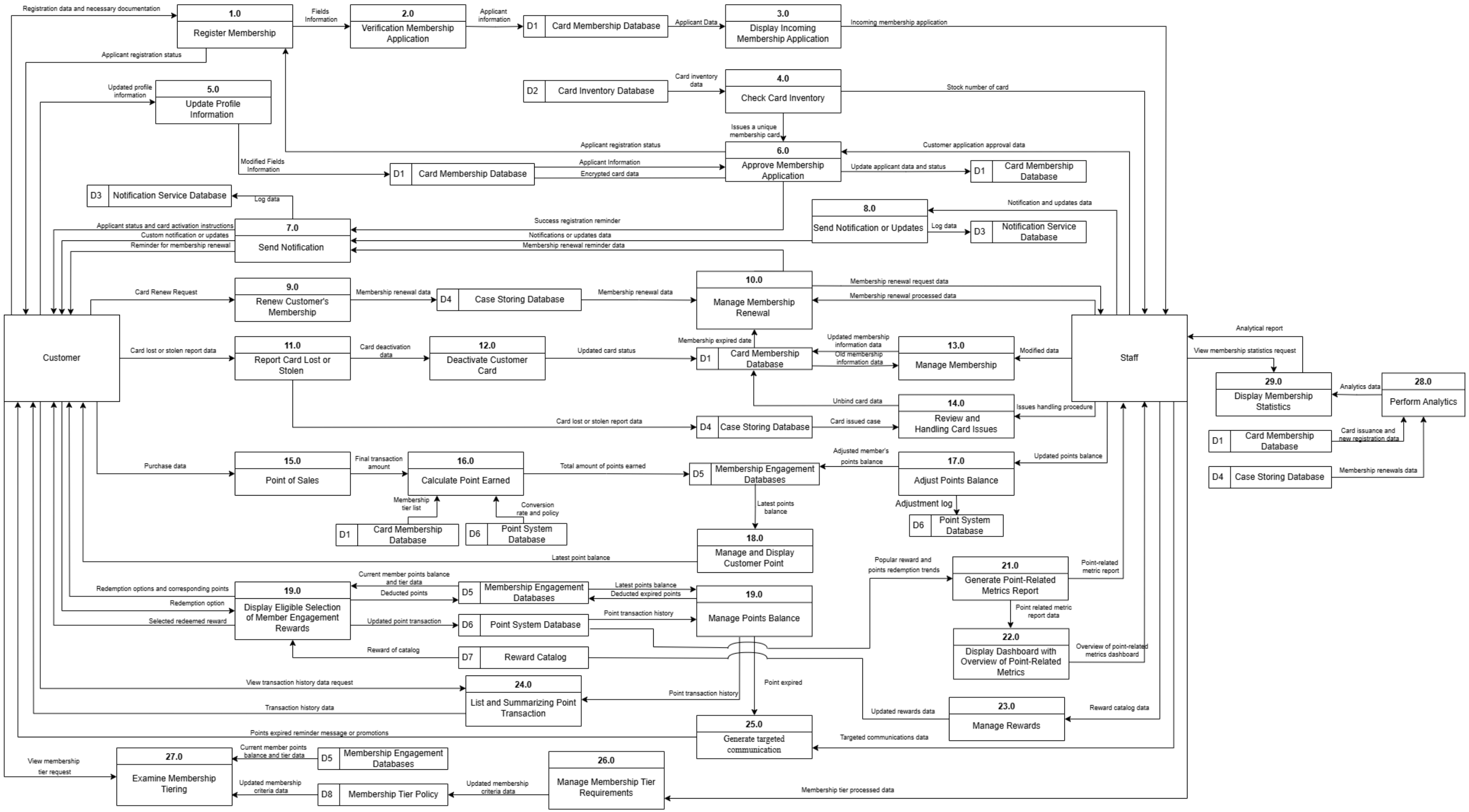


Figure 2.3.1.1: Digitally Drawn Membership Card Management System Data Flow Diagram Level 0

2.3.2 Use Case Diagram

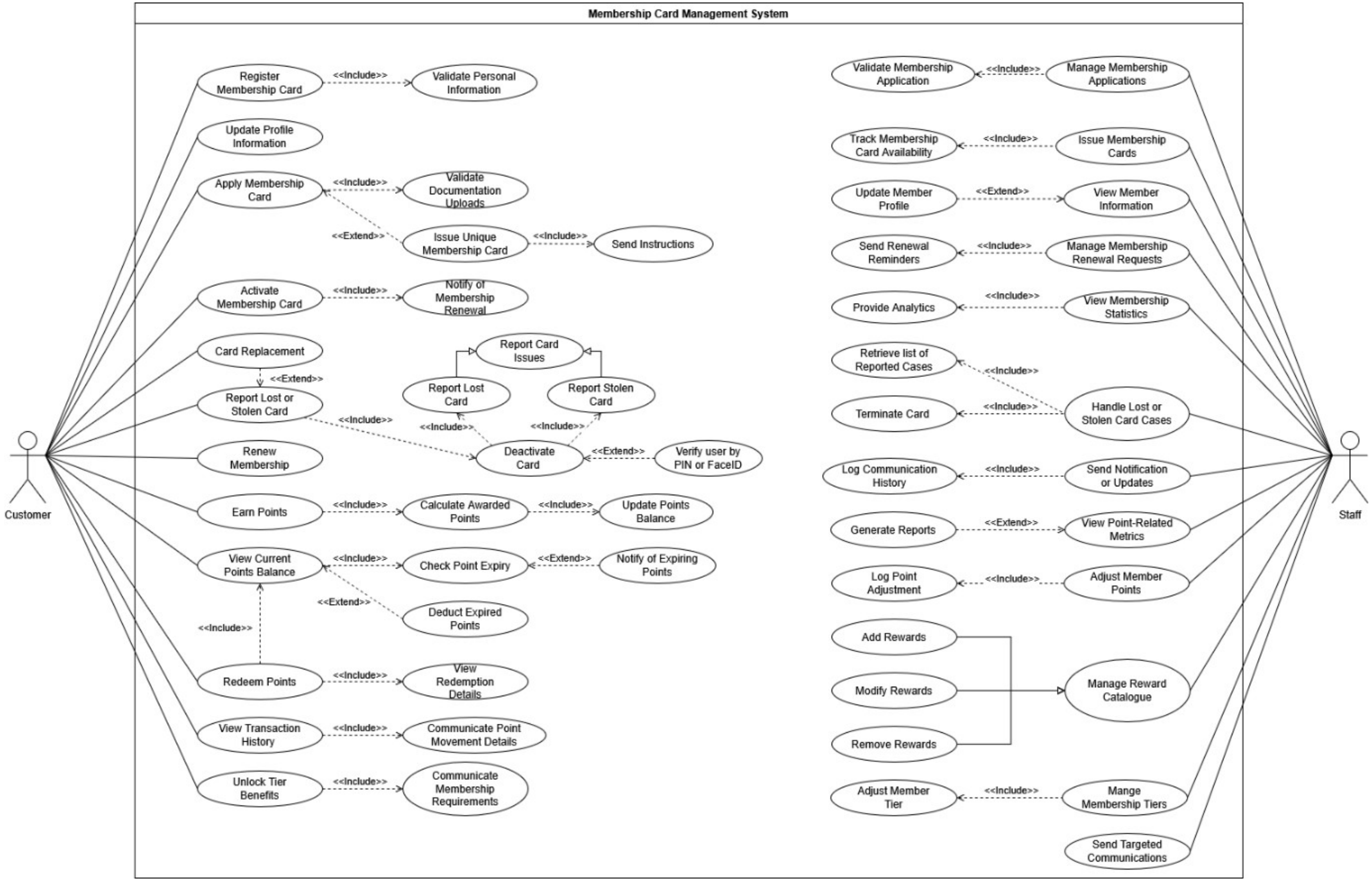


Figure 2.3.2.1: Digitally Drawn System's Functionalities for The Membership Card Management System Use Case Diagram

2.4 Use Case Description

2.4.1 Use Case Description

The system shall allow customers utilize membership points for item(s) redemption.

<table>
<tr><td>Use Case Name: Utilise Membership points for item(s) redemption</td><td>ID: 1</td><td>Importance Level: High</td></tr>
<tr><td>Primary Actor: Customer</td><td colspan="2">Use Case Type: Details, Essential</td></tr>
<tr><td colspan="3">Stakeholders and Interests:
Customer – wants to view current point balance, earn points, and redeem points for items
Staff – wants to adjust member points, and manage reward catalogue</td></tr>
<tr><td colspan="3">Brief Description:
This use case describes how we will handle the process that lets members able to utilise their membership points for item(s) redemption based on the available rewards in the reward catalogue managed by staffs.</td></tr>
<tr><td colspan="3">Trigger: Customers want to use their membership points to redeem items
Type: External</td></tr>
<tr><td colspan="3">Relationships:
Association: Customer
Include: View Current Point Balance, View Redemption Details
Extend: -
Generalisation: -</td></tr>
<tr><td colspan="3">Normal Flow of Events:
1. The customer signs in to the system.
2. The system validates if the member is a valid member.
3. The customer views their membership points.
4. The customer browses the items available for redemption.
5. The customer chooses an item to redeem.
6. The system shows the details of the item.
7. The system checks if the customer has sufficient membership points.
a. If the customer has sufficient points.
i. S-1: Subflow “Sufficient Points” is executed</td></tr>
<tr><td colspan="3">Subflows:
S-1: Sufficient Points
1. The customer confirms their decision to redeem the item.
2. The system deducts the points from the customer’s point balance.</td></tr>
</table>

2.4.2 Use Case Description

<table>
<tr><td>Use Case Name: Customers renew or reactivate their memberships.</td><td>ID: 2</td><td>Importance Level: High</td></tr>
<tr><td>Primary Actor: Customer</td><td colspan="2">Use Case Type: Details, Essential</td></tr>
<tr><td colspan="3">Stakeholders and Interests:
Customer – want to renew or reactivate their memberships and view the membership’s expiry date.
Staff – wants to handle membership renewal or reactivation request and ensure membership’s records are accurately updated.</td></tr>
<tr><td colspan="3">Brief Description:
This use case describes how we handle the process of for the members to renew or reactive their membership by providing the payment method and verifying their identities. It also outlines the staff's role in reviewing and approving the customer’s renewal or reactivation membership request.</td></tr>
<tr><td colspan="3">Trigger: Customer wants to renew or reactivate their membership
Type: External</td></tr>
<tr><td colspan="3">Relationships:
Association: Customer, Staff
Include: Validate login details, Check membership’s status, Verify customer credentials, Send renewal reminder, Update membership’s records
Extend: Process payment
Generalisation: Payment method</td></tr>
<tr><td colspan="3">Normal Flow of Events:
1. The customer login to their account.
2. The system validates the customer logins details
3. The customer navigates to the purchases and membership interface under settings.
4. The system retrieves and displays the membership expiry date.
5. The system checks the customer’s membership status:
a. If customer’s membership status is expired, the system displays the “Renew Membership” option
b. If customer’s membership status is suspended, the system displays the “Reactivate Membership” option
6. The customer selects the appropriate options.
7. The system prompts the customer to sign in again to authenticate their identity.
8. The system verifies the customer’s credentials.
9. The customer make payment by selecting a payment method and providing payment</td></tr>
</table>

3. The customer provides the necessary payment details. 4. The system validates payment details and processes the payment.
Alternate/Exceptional Flows: 2a. If the login details are invalid, the system will prompt the customer to enter their credentials again. 5a. If the sign in credentials is invalid, the customer does not allow to access to the renewal or reactivation membership process. 8a. If the payment transaction fails, the system does not proceed the renewal or reactivation membership process and notifies the customer to remake the payment.

Table 2.4.2.1: Use Case Description

2.4.3 Use Case Description [*3 – Yee Jia Hao]

<table>
<tr><td>Use Case Name: Customers reporting lost or stolen cards and deactivating the cards.</td><td>ID: 3</td><td>Importance Level: High</td></tr>
<tr><td>Primary Actor: Customer</td><td colspan="2">Use Case Type: Details, Essential</td></tr>
<tr><td colspan="3">Stakeholders and Interests:
Cardholder – want a direct way to report and deactivate their lost or stolen card as soon as possible to prevent unauthorized use.
Staff – want to receive urgent case reporting from customer to ensure efficient handling and to analyse potential card misuse.</td></tr>
<tr><td colspan="3">Brief Description:
This use case describes how we handle the process of customer’s request to report for lost or stolen card and deactivate it. This ensures the system provide an efficient automated card deactivation triggered by customer to prevent unauthorized use of cardholder’s card as fast as possible and provide crucial support to customer immediately.</td></tr>
<tr><td colspan="3">Trigger: User report lost or stolen card
Type: External</td></tr>
<tr><td colspan="3">Relationships:
Association:
Customer is associated with Report lost or stolen card
Staff is associated with Handle lost or stolen card cases
Include:
Report Lost or Stolen Card –(i)–> Deactivate Card
Deactivate Card –(i)–> Report Lost Card
Deactivate Card –(i)–> Report Stolen Card
Handle Lost or Stolen Card Cases –(i)–> Retrieve Reported Cases
Handle Lost or Stolen Card Cases –(i)–> Terminate Card
Extend:
Card Replacement –(e)–> Report Lost or Stolen Card
Verify user by PIN or FaceID –(e)–> Deactivate Card
Generalisation:
Report Lost Card and Report Stolen Card are generalized as Create Card Case</td></tr>
<tr><td colspan="3">Normal Flow of Events:
1. Customer enter the card management user interface.
2. Customer select lost or stolen card option.
3. System verifies the user by PIN or Face ID.</td></tr>
</table>

Subflows: **S-1: Card Replacement** 1. System redirect customer to card replacement interface to replace the lost or stolen card. 2. Customer enter the new card name and card delivery location. 3. Customer make payment. 4. System process payment. 5. System processes the order placement for replacement card. 6. Staff handle the replacement card delivery to their customer. 7. Customer obtain and activate their replacement card.
Alternate/Exceptional Flows: 3a. User failed to verify their identity for 3 times, system will advise the customer to report and deactivate their lost or stolen card through customer service channel. S-1: 4a. If payment transaction failed, prompt user to retry the payment process.

*Table 2.4.3.1: Use Case Description [*3 – Yee Jia Hao]*

2.4.4 Use Case Description

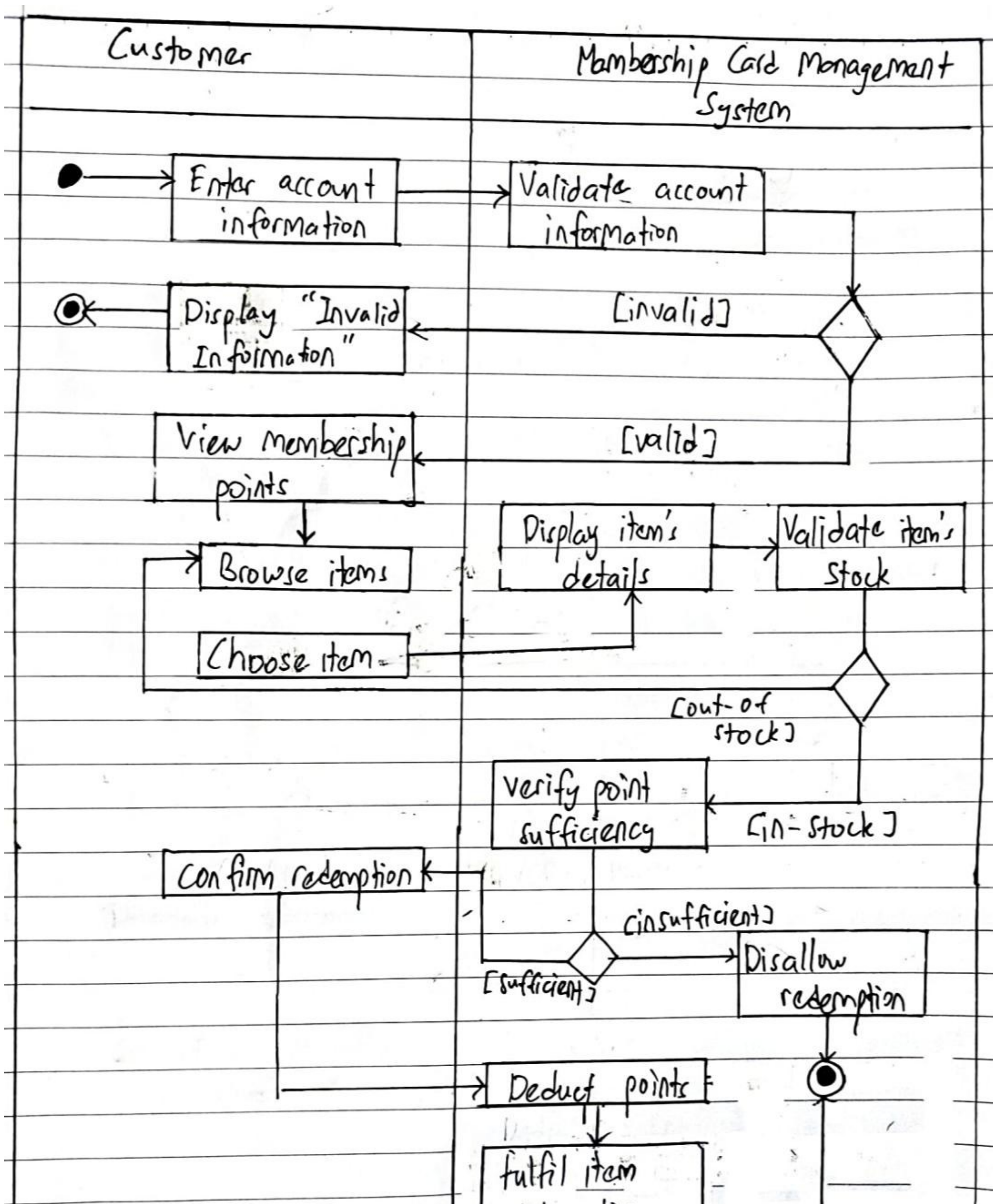
<table>
<tr><td>Use Case Name: Customers seek to return and refund item(s) redemption.</td><td>ID: 4</td><td>Importance Level: High</td></tr>
<tr><td>Primary Actor: Customer</td><td colspan="2">Use Case Type: Details, Essential</td></tr>
<tr><td colspan="3">Stakeholders and Interests:
Customer- want to return and refund item(s) redemption
Staff- manage return and refund of items redemption</td></tr>
<tr><td colspan="3">Brief Description: This use case describes how a customer can request a return and refund for an incorrect or damaged item.</td></tr>
<tr><td colspan="3">Trigger: Customers initiate return and refund of item(s) request.
Type: External</td></tr>
<tr><td colspan="3">Relationships:
Association: Customer
Include: Seek return and refund items redemption
Extend: -
Generalisation: -</td></tr>
<tr><td colspan="3">Normal Flow of Events:
1. Customer login to their account.
2. Customer selects the item they want to return.
3. Customer initiates return request.
4. System validates allowed return period.
5. Customer selects the reason for the return.
6. Customer uploads supporting evidence like photos or videos.
7. Customer submits the return request.
8. Staff review the return request.
a. If the request is valid.
i. S-1: Return approval process in performed
9. Customer selects the return option.
a. If the customer selects drop-off return option.
i. S-2: Drop-off process in performed.
b. If the customer selects pickup return option.
i. S-3: Pickup process in performed.
10. System sends tracking updates to customer.
11. System make confirmation once receive the returned items.</td></tr>
</table>

2. Logistics partner picks up the item from the customer's address.
Alternate/Exceptional Flows: 4a1: If the customer returns an item after the allowed return period, reject the request. 4a2: System notifies the customer of an invalid request. 8a1: If the evidence submitted by the customer does not satisfy the return criteria, reject the request. 8a2: System notifies the customer of an invalid request.

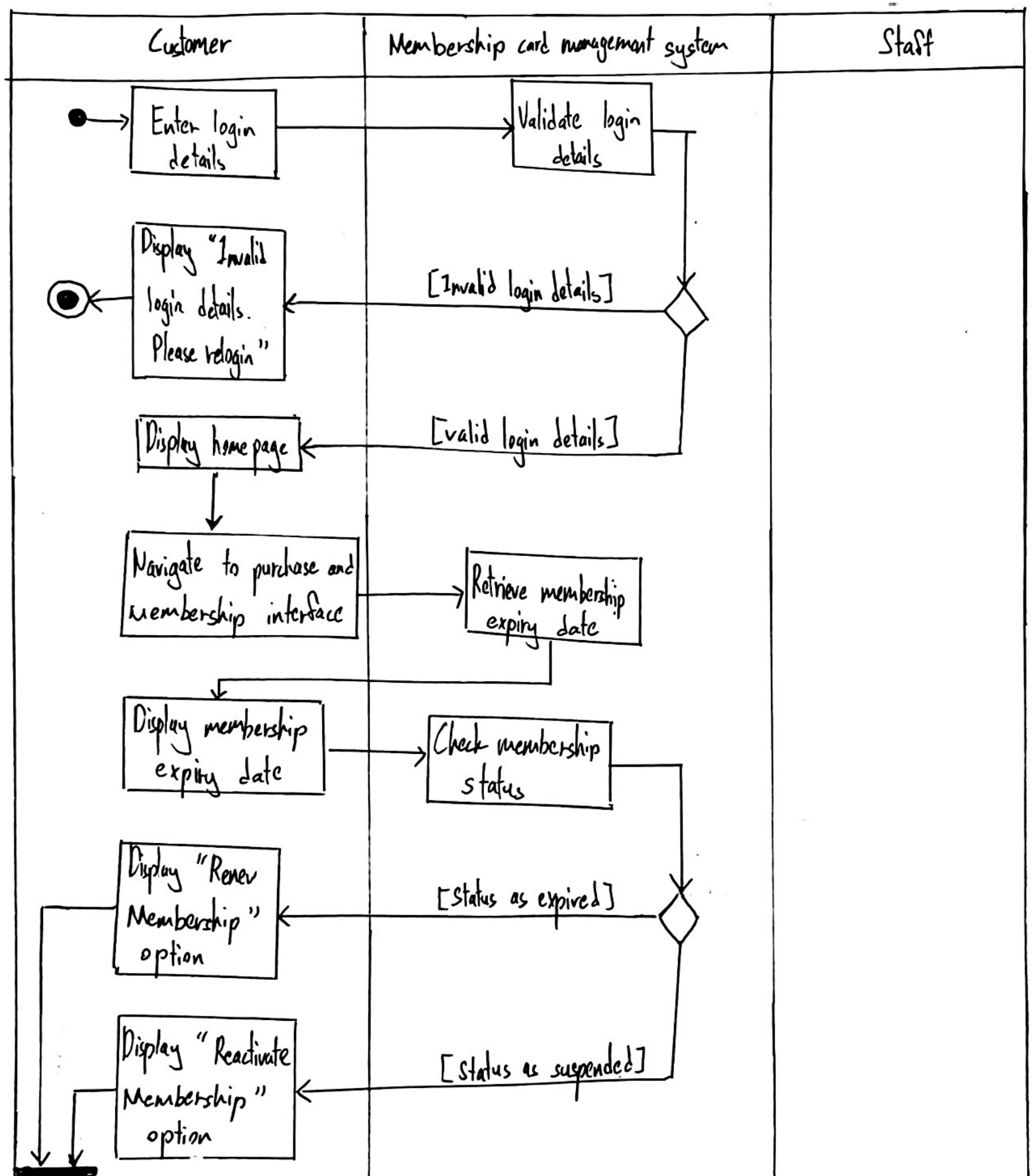
Table 2.4.4.1: Use Case Description

2.5 Activity Diagrams

2.5.1 Activity Diagrams



2.5.2 Activity Diagrams



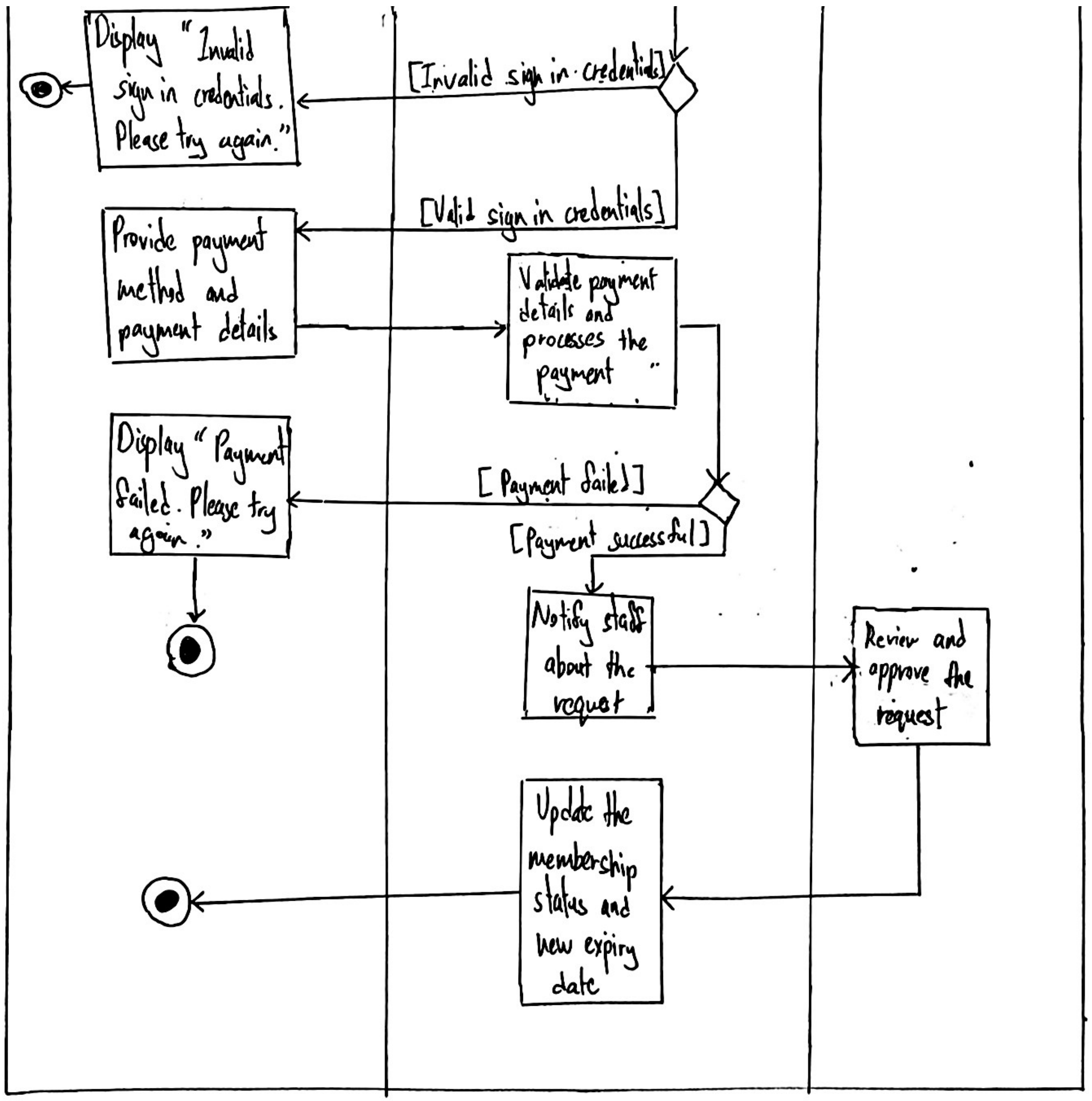
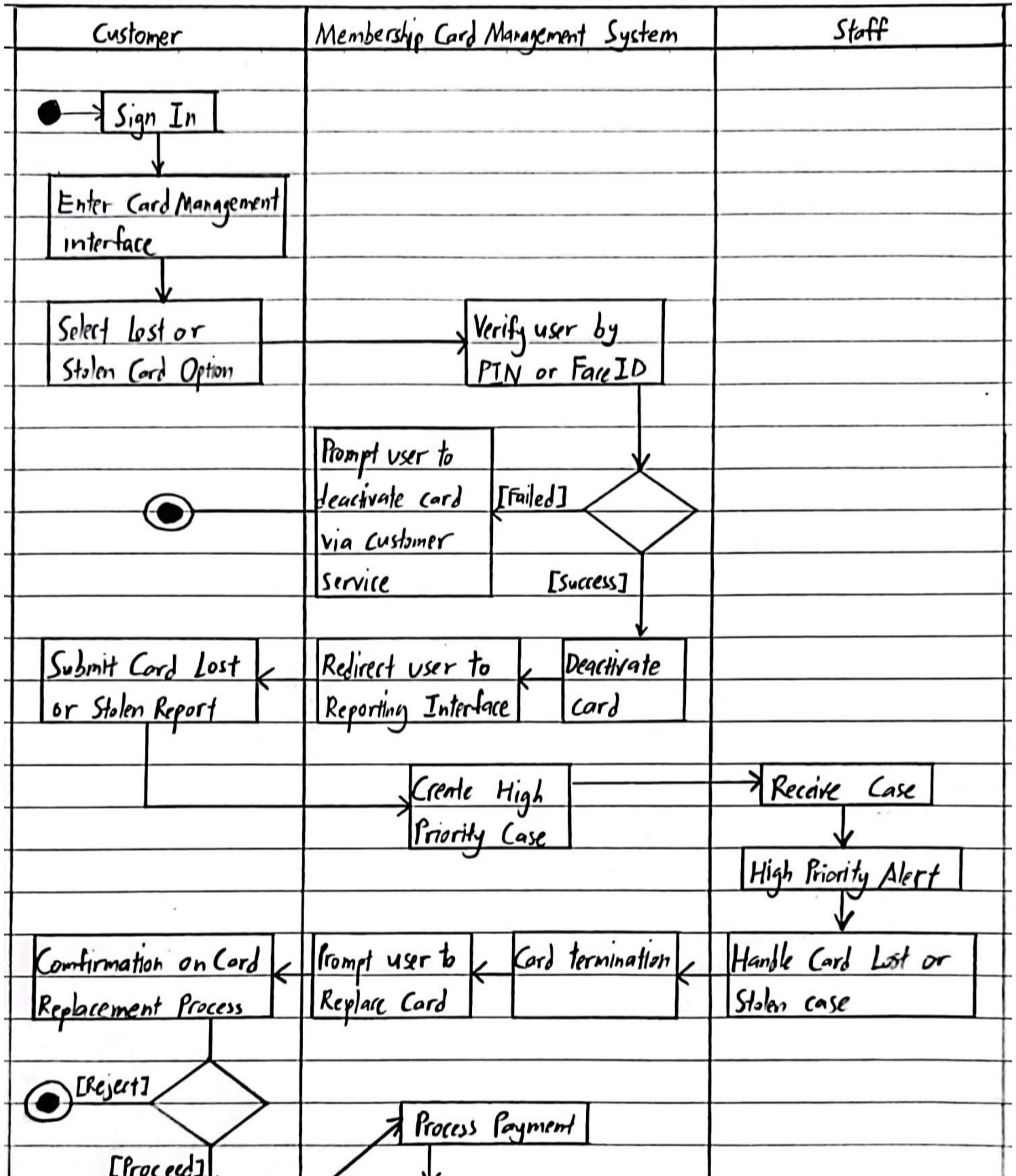
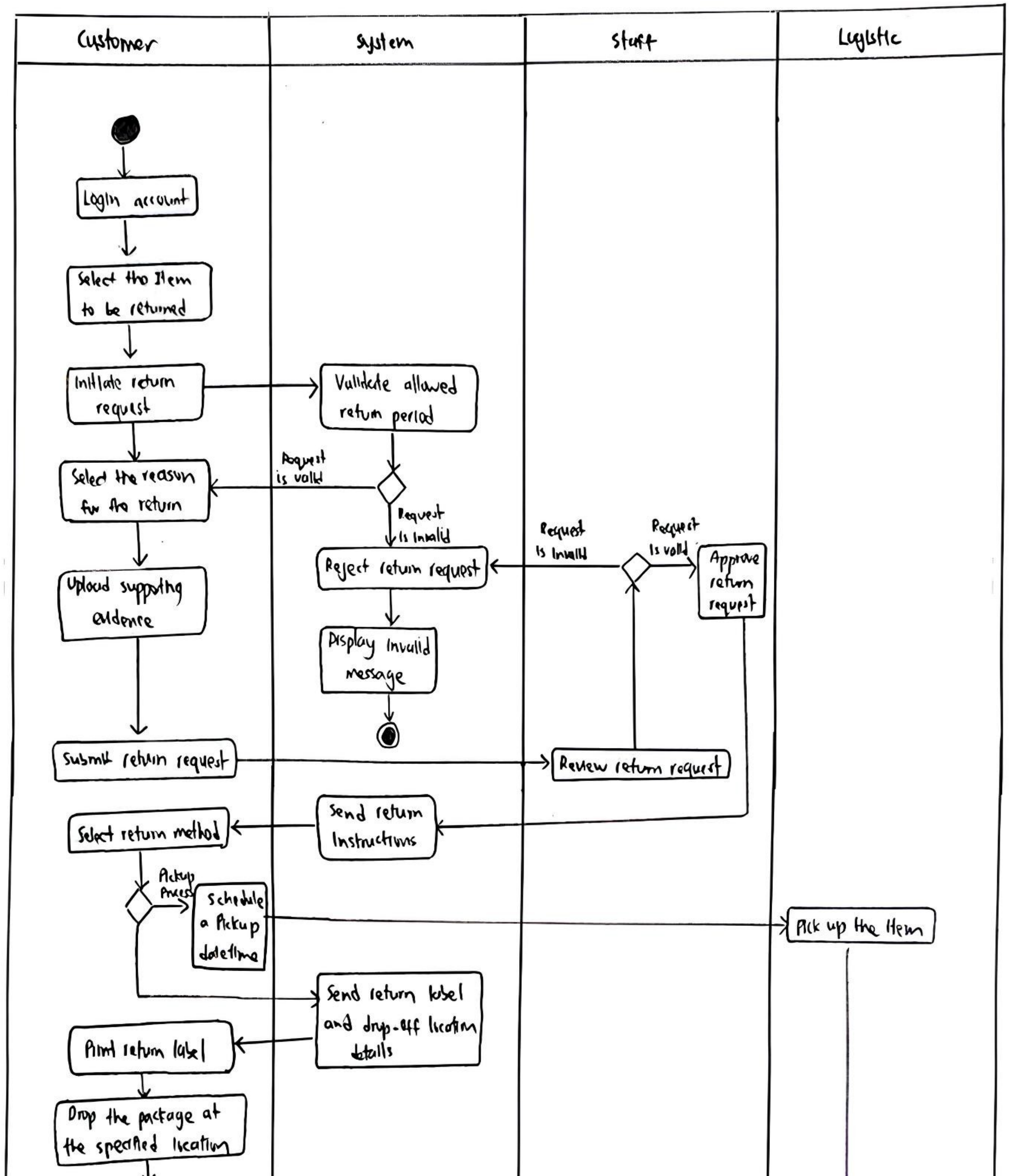


Figure 2.5.2.1: Activity Diagrams

2.5.3 Activity Diagrams



2.5.4 Activity Diagrams



2.6 Class Diagram